What does the Mediterranean diet actually do for health — and how strong is the evidence?

ABSTRACT — The Mediterranean diet is the most studied dietary pattern in medicine, and its benefits are real but mostly modest. The evidence is firmest for the heart: randomised trials in three different settings — primary prevention, and twice in people with existing coronary disease — show roughly 20–30% fewer major cardiovascular events, and large cohort studies on three continents consistently link higher adherence to about 10–25% lower death rates, with a dose–response gradient. Diabetes prevention is the outcome where trial and observational evidence agree best, supported by an objective blood-biomarker check. Beyond that, the picture softens: effects on depression, liver fat, weight, and some cancers are promising but rest on small or observational studies, and for cognition the cohort studies say yes while the largest dedicated trial said no. The evidence base has known soft spots. Most of it is observational and confounded by the healthier lives of adherent eaters; effects shrink outside Mediterranean countries; trial benefits depend on intensive support; and the flagship trial had to be retracted and republished after randomisation failures — though the corrected estimate barely moved. A fair summary is that the diet probably prevents cardiovascular events and diabetes, is associated with longer life at modest effect size, and carries plausible, partly demonstrated mechanisms, while several of its most publicised benefits remain unproven at trial standard.

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Introduction

Few ideas in nutrition have travelled further than the claim that the traditional diet of the olive-growing Mediterranean protects against chronic disease. The claim now spans mortality, cardiovascular disease, diabetes, cancer, dementia, depression, liver disease, and frailty, and an umbrella review counts robust evidence for some of these and only suggestive evidence for others Dinu et al. 2018. The pattern is also unusual among diets in having hard-endpoint randomised trials at all, which makes it a rare case where the observational story can be checked against experiment Estruch et al. 2018, de Lorgeril et al. 1999, Delgado-Lista et al. 2022.

The question deserves a sceptical reading as much as an enthusiastic one. Nearly all of the mortality evidence is observational — a Dietary Guidelines systematic review found 152 cohort-type studies and a single randomised trial — and the field's flagship trial was retracted and republished after problems with its randomisation came to light English et al. 2021, Estruch et al. 2018. This review therefore works outcome by outcome, from the best-supported claims to the weakest, giving the contrary evidence its own space: what the diet is, where the claim came from, what trials and cohorts show for each outcome, what mechanisms have actually been measured, and why the headline numbers may still flatter the diet.

Definition and measurement

"Mediterranean diet" names a scored pattern, not one menu. Research definitions converge on abundant vegetables, fruit, legumes, and unrefined cereals; olive oil as the principal fat, giving a high ratio of monounsaturated to saturated fat; regular fish; little red or processed meat; moderate dairy; and moderate wine with meals Bach et al. 2006, Menotti & Puddu 2025. No single food defines the pattern; the score rises with the overall profile.

Adherence is measured with a priori scores rather than by asking whether someone "eats Mediterranean". The original 0–8/0–9 Trichopoulou scale and its descendants dominate the cohort literature, while trials mostly use the 14-item Mediterranean Diet Adherence Screener (MEDAS) Trichopoulou et al. 1995, Bach et al. 2006. Dozens of score variants now exist, with different components and cut-offs, and this variety complicates every meta-analysis in the field, since a "2-point increment" does not mean quite the same thing from one study to the next Bach et al. 2006, Dinu et al. 2018.

Origins: ecology before trials

The health claim began as ecology, not experiment. In the Seven Countries Study — 15 cohorts, 11,579 men, 15 years of mortality follow-up — coronary death rates were lowest in the cohorts where olive oil was the main dietary fat, and the monounsaturated-to-saturated fat ratio tracked both all-cause and coronary mortality across cohorts Keys et al. 1986. This was an ecological association across populations, and its authors made no causal claim Keys et al. 1986.

That ecology nonetheless set the agenda for half a century. Ancel Keys' books and the study's cohort contrasts effectively defined the diet as a research object Menotti & Puddu 2025, Blackburn 2017. Later re-analyses of the study's food data produced adherence scores whose ecological correlation with 50-year coronary mortality across cohorts reached −0.91 — still ecology, but remarkably consistent ecology, and a reminder that the modern scores were built to formalise an old observation rather than discover a new one Menotti & Puddu 2025.

Cardiovascular outcomes: the trials

The strongest claim for the diet is that it prevents cardiovascular events, and randomised trials support it in three different settings. In primary prevention, the PREDIMED trial randomised 7,447 adults at high cardiovascular risk to a Mediterranean diet supplemented with extra-virgin olive oil, the same diet supplemented with nuts, or low-fat advice; over 4.8 years, major cardiovascular events occurred in 3.8% and 3.4% of the two Mediterranean arms against 4.4% of controls, hazard ratios 0.69 (95% CI 0.53–0.91) and 0.72 (0.54–0.95) Estruch et al. 2018.

Secondary prevention came first historically, and with a much larger apparent effect. The Lyon Diet Heart Study randomised 605 patients after myocardial infarction and, over 46 months, recorded cardiac death plus nonfatal infarction in 14 patients on the Mediterranean-style diet against 44 on a prudent Western diet, adjusted risk ratios 0.28–0.53 de Lorgeril et al. 1999. An effect of that magnitude has never been replicated, and it is best read as an early outlier rather than the expected yield of the diet de Lorgeril et al. 1999.

The modern secondary-prevention test lands between the two. CORDIOPREV followed 1,002 patients with coronary disease for seven years and found 87 major events on the Mediterranean diet against 111 on a low-fat diet — 28.1 versus 37.7 events per 1,000 person-years, adjusted hazard ratios 0.72–0.75 Delgado-Lista et al. 2022. The benefit was evident in men but not demonstrated in the trial's 175 women, a subgroup too small to settle the question either way Delgado-Lista et al. 2022.

Pooled across randomised trials, the pattern holds for events but not for deaths: composite cardiovascular events fell (RR 0.63, 95% CI 0.53–0.75) while all-cause mortality did not move (RR 1.00, 0.86–1.15) Liyanage et al. 2016. Table 1 lines up the landmark trials, including two — DIRECT on weight and the MIND trial on cognition — that anchor later sections.

TRIAL	SETTING, N	COMPARATOR	MAI RES (95%
Lyon Diet Heart (1999)	Post-MI, 605	Prudent Western diet	Card deat 14 v even adju 0.28
PREDIMED (2018 republication)	Primary prevention, high risk, 7,447	Low-fat advice	MAC 0.69 0.91 oil; ((0.5 nuts
CORDIOPREV (2022)	Coronary disease, 1,002	Low-fat diet	MAC 0.72 28.1 even pers
DIRECT (2008)	Obesity, 322	Low-fat / low-carb	Weig kg (l −2.9 fat) (low 2 y
MIND diet trial (2023)	Cognition, older adults, 604	Calorie-matched usual diet	Glob cogn diffe 0.03 (−0. 0.09

The PREDIMED retraction

The field's flagship trial did not survive scrutiny intact, and what happened next is part of the evidence. A statistical screen of baseline data across 5,087 trials flagged improbable distributions, which prompted close examination of PREDIMED's randomisation Carlisle 2017. The investigators then found protocol deviations — household members enrolled without individual randomisation, and randomisation misapplied at two of the trial's eleven sites — withdrew the 2013 report, and republished the trial in 2018 with analyses that no longer assumed full randomisation Estruch et al. 2018.

The corrected estimate barely moved. Omitting the 1,588 participants with known or suspected departures from randomisation left the hazard ratios essentially unchanged, at 0.69 and 0.72 for the two Mediterranean arms Estruch et al. 2018. The episode is therefore double-edged: it exposed real conduct failures in the field's central trial, and it also showed the headline result to be robust to their correction.

Two caveats survive the correction and should temper how the trial is quoted. The composite result was driven mainly by stroke — hazard ratio 0.60 (95% CI 0.45–0.80), roughly 24 down to 14 strokes per 1,000 — rather than by infarction or death Rees et al. 2019. And PREDIMED showed no effect on total mortality (HR 1.0, 0.81–1.24), an outcome for which it was not powered; the trial demonstrates fewer events, not longer lives Guasch-Ferré et al. 2017, Rees et al. 2019.

What the syntheses conclude

Independent syntheses back the cardiovascular benefit at moderate certainty. A network meta-analysis of 40 trials with 35,548 patients rated Mediterranean dietary programmes superior to minimal intervention with moderate GRADE certainty: all-cause mortality OR 0.72 (95% CI 0.56–0.92), cardiovascular mortality 0.55 (0.39–0.78), stroke 0.65 (0.46–0.93), and nonfatal infarction 0.48 (0.36–0.65), amounting to about 17 fewer deaths per 1,000 over five years at intermediate risk Karam et al. 2023. An umbrella review of trial meta-analyses put cardiovascular risk ratios in the same neighbourhood, at 0.55–0.64 Chareonrungrueangchai et al. 2020.

Against this, the comparator matters more than promotional summaries admit. The same network meta-analysis found no convincing advantage of Mediterranean over low-fat programmes for mortality or infarction — the diet clearly beats doing nothing, but not clearly the main alternative Karam et al. 2023. The Cochrane review is cooler still: across 30 randomised trials and 12,461 participants it concludes that "some uncertainty" remains, with risk-factor effects that are small — systolic blood pressure −3.0 mmHg (95% CI −3.5 to −2.5), total cholesterol −0.16 mmol/L (−0.32 to 0.00) — and evidence quality low to moderate throughout Rees et al. 2019. The honest reading is a real benefit of moderate certainty against minimal intervention, an unsettled contest against other structured diets, and unimpressive movement in classical risk factors.

Mortality in the cohorts

Large cohort studies agree on lower mortality with higher adherence, and they show a dose–response. In the founding cohort of 22,043 Greek adults, each 2-point increment on the adherence score predicted 25% lower total mortality (HR 0.75, 95% CI 0.64–0.87), and across 74,607 elderly Europeans the same increment predicted an 8% reduction (3–12%) Trichopoulou et al. 2003, Trichopoulou et al. 2005. The finding replicates outside Europe: in the Nurses' Health Study and Health Professionals cohorts, top-quintile Alternate Mediterranean scores predicted mortality HR 0.82 (0.79–0.84) over up to 36 years and cardiovascular disease HR 0.83 (0.79–0.86), and in 25,315 US women followed up to 25 years the top-adherence hazard ratio was 0.77 (0.70–0.84) Shan et al. 2023, Shan et al. 2020, Ahmad et al. 2024.

The pooled effect, however, is smaller than the early Greek numbers suggested. Meta-analyses of cohorts converge near HR 0.90 per 2 points (0.89–0.91; 29 studies, 1.68 million people) and 0.79 (0.77–0.81) for highest-versus-lowest adherence, and the newest synthesis of 54 cohorts and 1.83 million people grades the evidence moderate at RR 0.96 (0.95–0.97) per single point Sofi et al. 2008, Soltani et al. 2019, Eleftheriou et al. 2018, Nucci et al. 2026. Shrinkage from HR 0.75 to about 0.90 per 2 points as the literature matured is itself informative: the association is consistent, but its size in the founding cohort was not the size in the world.

The association extends to groups where it matters most, and it is not carried by any single food. It holds in older adults (all-cause mortality RR 0.77, 0.70–0.83) and in people who already have diabetes (2-point HR 0.63; pooled RR 0.93, 0.90–0.97) Furbatto et al. 2024, Bonaccio et al. 2016, Muscogiuri et al. 2026. Component analyses attribute most of the mortality association to moderate alcohol (~24%), low meat (~17%), and high vegetable intake (~16%), with cereals, dairy, and fish contributing little — evidence for a pattern effect rather than a "superfood" effect Trichopoulou et al. 2009.

Type 2 diabetes

Diabetes is where trial and cohort evidence line up best. Within PREDIMED, the pooled Mediterranean arms cut incident diabetes (HR 0.70, 95% CI 0.54–0.92), and yearly-measured adherence showed a gradient — HR 0.80 (0.70–0.92) per 2 points and 0.46 (0.25–0.83) for highest versus lowest adherence Bloomfield et al. 2016, Martínez-González et al. 2023. Adding energy restriction and exercise helps further: PREDIMED-Plus, with 4,746 adults over six years, cut diabetes incidence an additional 31% (95% CI 18–41%) relative to an unrestricted Mediterranean diet, an absolute risk of 9.5% versus 12.0% Ruiz-Canela et al. 2025.

The diet also improves established diabetes. Meta-analyses of randomised trials show better glycaemic control, with HbA1c reductions of 0.30–0.47 percentage points and BMI −0.83 (95% CI −1.40 to −0.26), and long-term trials show a 49% higher probability of remission from the metabolic syndrome Esposito et al. 2015, Wu et al. 2025.

Two independent checks argue that this signal is not an artefact of self-report. A blood biomarker score of Mediterranean eating — carotenoids plus fatty acids, validated in a feeding trial — predicted 29% lower diabetes incidence per standard deviation (HR 0.71, 0.65–0.77) in 22,202 EPIC-InterAct participants, a larger effect than self-reported scores suggest Sobiecki et al. 2023. And in the decidedly non-Mediterranean UK Biobank, a Mediterranean lifestyle score predicted 30% lower diabetes incidence (top-quartile HR 0.70, 0.62–0.79), showing the association travels Maroto-Rodriguez et al. 2023.

Risk factors, weight, and body composition

Cardiometabolic risk factors improve consistently on the diet, but modestly. Meta-analyses of trials and cohorts show small shifts across the metabolic syndrome's components — waist circumference −0.42 cm, triglycerides −6.1 mg/dl, systolic blood pressure −2.4 mmHg, glucose −3.9 mg/dl — and a pooled metabolic syndrome risk ratio around 0.81 Kastorini et al. 2011, Godos et al. 2017, Papadaki et al. 2020. None of these numbers would impress a pharmacologist; their claim to importance is that they arrive together and persist.

For weight, the diet is competitive rather than superior — but unusually durable. In the 2-year DIRECT trial it beat low fat (−4.4 vs −2.9 kg) and matched low carbohydrate (−4.7 kg), and a network meta-analysis found it the only named diet whose cardiovascular risk-factor benefits persisted at 12 months Shai et al. 2008, Ge et al. 2020. Body-composition trials point the same way: an energy-restricted Mediterranean diet plus activity reduced total and visceral fat and attenuated lean-mass loss over three years, and a polyphenol-enriched "green" variant more than doubled visceral-fat loss (−14.1% vs

−6.0%) Konieczna et al. 2023, Zelicha et al. 2022.

The contrary case sits close by. A 2024 meta-analysis of 18 trials of Mediterranean diet enriched with olive oil found no consistent benefit on any cardiometabolic marker except triglycerides (−2.4 mg/dl, 95% CI −4.5 to −0.3), and a GRADE-based review in overweight adults rated most of the risk-factor evidence very low quality Keshani et al. 2024, Hernandez et al. 2025. Supplementing one component, in short, is not the same as changing the pattern — and even the pattern's risk-factor effects rest on shakier ground than its event reductions.

Cancer

The observational cancer signal is broad but shallow. Cohort meta-analyses associate the highest adherence with lower cancer mortality (RR 0.86–0.87) and lower colorectal cancer incidence (0.82–0.83), with a weaker association for breast cancer (0.94, 95% CI 0.90–0.97) Morze et al. 2021, Schwingshackl et al. 2017. An umbrella review rates the evidence robust only for overall cancer incidence and mortality; for most site-specific cancers it is suggestive or weak, and for several sites null Dinu et al. 2018, Morze et al. 2021.

Trial evidence amounts to a single fragile signal. PREDIMED's olive-oil arm recorded fewer invasive breast cancers (HR 0.32, 95% CI 0.13–0.79), but this was a secondary outcome with only 35 cases in total, and it needs confirmation before it carries weight Toledo et al. 2015. Olive oil itself shows the same asymmetry: at +25 g/day it is associated with lower cardiovascular disease, diabetes, and mortality across 1.29 million participants, but not with lower cancer risk (RR 0.94, 0.86–1.03) Martínez-González et al. 2022. For cancer, the fair verdict is an association worth having, with no demonstrated causal protection for any specific site.

Cognition

Cognition is the sharpest trial–cohort disagreement in the field. The cohorts and their meta-analyses are consistently positive: cognitive impairment HR 0.82 (0.75–0.89), dementia 0.89 (0.83–0.95), and Alzheimer's disease 0.70 (0.60–0.82) Fekete et al. 2025, Fu et al. 2022. In 60,298 UK Biobank participants, top-tertile adherence predicted 23% lower dementia risk (HR 0.77, 0.65–0.91) independent of polygenic risk, and the related MIND diet shows a pooled dementia hazard ratio of 0.83 (0.76–0.90) across cohorts Shannon et al. 2023, Chen et al. 2023.

Early trial evidence appeared to agree. PREDIMED's Navarra sub-study found better MMSE scores after 6.5 years on the olive-oil diet (+0.62 points, 95% CI 0.18–1.05, versus control), and pooled randomised trials show an episodic-memory benefit of SMD 0.20 (0.09–0.30) Martínez-Lapiscina et al. 2013, Fu et al. 2022.

Then the largest dedicated trial came back null. The 3-year MIND trial randomised 604 older adults to the MIND diet or a calorie-matched usual diet, both with mild calorie restriction, and found a global-cognition difference of 0.035 SD (95% CI −0.022 to 0.092, p=0.23) and no differences on MRI; both groups improved and both lost about 5 kg Barnes et al. 2023. The likeliest reconciliations are an unusually active, weight-losing control arm, a well-nourished volunteer population, and residual confounding in the cohorts Barnes et al. 2023, Maki et al. 2014. The trial cannot rule out effects over longer durations or in higher-risk populations, but it caps what can be claimed for three years of dietary change in healthy older volunteers Barnes et al. 2023.

Depression

Depression is the outcome where small trials report the largest effects — and where the synthesis urges the most caution. Observationally, higher adherence is associated with 33% lower incident depression (OR 0.67, 95% CI 0.55–0.82) Lassale et al. 2019. Pooling five randomised trials with 1,507 participants, depressive symptoms fell by SMD −0.53 (95% CI −0.90 to −0.16) — a moderate effect by convention, and larger than most drug–placebo differences in mild depression Bizzozero-Peroni et al. 2025. Table 2 shows the three trials that drive it.

TRIAL	N, DURATION	CONTROL	RESULT
SMILES (2017)	67 adults, major depression, 12 wk	Social support	MADRS −7.1 points vs control; remission 32% vs 8%; NNT 4.1
HELFIMED (2019)	152 adults, self-reported depression, 3–6 mo	Social groups	Greater depression reduction at 3 mo, sustained at 6 mo
AMMEND (2022)	72 young men, moderate–severe depression, 12 wk	Befriending	BDI-II improved 14.4 points more (95% CI 11.4–17.4)

The caution is structural, not cosmetic. Heterogeneity across the pooled trials is high ($I^2 = 87\%$), the prediction interval spans zero (−1.86 to 0.81), and the certainty of evidence is rated low Bizzozero-Peroni et al. 2025. All three positive trials are small, unblinded to participants, and use comparators — befriending, social groups — that may underperform genuine usual care, which would inflate the apparent benefit of the diet arm Bizzozero-Peroni et al. 2025. The direction is consistent and encouraging; the size is not yet trustworthy.

Liver, frailty, and joints

For fatty liver disease, small trials point one way and one dissents. A meta-analysis of six small randomised trials shows reduced steatosis (fatty-liver index SMD −1.06, 95% CI −1.95 to −0.17) and improved insulin resistance, and a crossover trial confirmed steatosis reduction even without weight loss Kawaguchi et al. 2021, Ryan et al. 2013. The 18-month DIRECT-PLUS trial went further, nearly doubling intrahepatic-fat loss with a green-Mediterranean variant (−38.9% versus −12.2% proportional loss on guideline care) Yaskolka Meir et al. 2021. Against this, one 12-week trial found no advantage over a low-fat diet, and clinical reviews still place calorie restriction, not diet composition, at the centre of NAFLD management George et al. 2022, Haigh et al. 2022.

The frailty and joint evidence is observational or thin, respectively. Higher adherence predicts roughly half the risk of incident frailty in Spanish cohorts (odds ratios 0.48 and 0.38 for top adherence) and lower frailty plus 8-year mortality in US NHANES participants León-Muñoz et al. 2014, Maroto-Rodriguez et al. 2022, Jayanama et al. 2021. In rheumatoid arthritis, a systematic review finds improvement in some disease-activity measures, smaller and less consistent than the effects of omega-3 supplementation Philippou et al. 2021. These are leads, not conclusions.

Mechanisms

The proposed mechanisms are measurable, not just plausible, and several have been demonstrated inside the hard-endpoint trials themselves. In CORDIOPREV, one year of the Mediterranean diet produced better endothelial function than low fat — flow-mediated dilation 3.83% (95% CI 2.91–4.23) versus 1.16% (0.80–1.98) — and by 5–7 years carotid intima–media thickness had regressed (−0.027 mm) while the low-fat arm's had not Yubero-Serrano et al. 2020, Jimenez-Torres et al. 2021. The diet, in other words, changes the artery wall in the same patients in whom it changes event rates.

Inflammation is the second measured pathway. Top-quintile Mediterranean scores predict roughly 24% lower C-reactive protein (CRP), and in the Women's Health Study, inflammatory and small-molecule metabolite biomarkers were the largest single mediators of the diet–mortality association Fung et al. 2005, Ahmad et al. 2024.

The newer mechanistic frontier runs through the gut and the biology of aging. One-year interventions across five countries shifted the gut microbiota toward short-chain-fatty-acid producers, tracking lower CRP and IL-17 and better frailty markers, and shorter trials confirm microbiome and bile-acid shifts alongside improved insulin sensitivity Ghosh et al. 2020, Meslier et al. 2020, Muralidharan et al. 2021, Rinott et al. 2022. The polyphenol-rich green-Mediterranean variant has, in addition, attenuated epigenetic-clock aging and age-related brain atrophy in participants over 50 Yaskolka Meir et al. 2023, Kaplan et al. 2022. Mechanism does not substitute for outcomes, but it makes the outcome data harder to dismiss as confounding all the way down.

Why the headline numbers may still flatter the diet

The deepest problem is that almost all of the mortality evidence is observational. The Dietary Guidelines review's count — 152 cohort-type studies against one randomised trial — describes a literature in which adherent eaters also exercise more, smoke less, and are richer, and no adjustment removes all of that English et al. 2021, Maki et al. 2014, Ioannidis 2018. Measurement compounds the confounding: self-reported food-frequency instruments are noisy, and the dozens of non-interchangeable adherence scores in circulation make effects hard to compare across studies Bach et al. 2006, Ioannidis 2018.

Generalisability is a second, quantified worry. The pooled mortality association is HR 0.82 per 2 points inside Mediterranean countries but 0.92 elsewhere — the effect measurably shrinks with distance from the Mediterranean, whether because of food quality, dose, or culture Soltani et al. 2019. Trial effects, meanwhile, depend on intensive support — free foods, dietitian counselling, quarterly group sessions — so unsupported real-world adoption will likely deliver less, and some trials are part-funded by olive-oil sector foundations Rees et al. 2019, Karam et al. 2023, Delgado-Lista et al. 2022.

Weighed together, the umbrella-review verdict is a fair summary: robust evidence for mortality, cardiovascular disease, overall cancer incidence, neurodegenerative disease, and diabetes; suggestive or weak evidence for most everything else Dinu et al. 2018. None of these caveats erases the cardiovascular trials; they mostly caution against transplanting cohort-sized expectations into non-Mediterranean kitchens.

Open questions

The next round of evidence is already designed. Hard-endpoint results from PREDIMED-Plus — 6,874 participants in an 8-year cardiovascular follow-up of energy-restricted Mediterranean diet plus exercise — will test whether adding weight loss amplifies the PREDIMED effect Konieczna et al. 2023, Ruiz-Canela et al. 2025. Objective adherence biomarkers, metabolomic and carotenoid/fatty-acid scores among them, can cut the reporting bias that plagues cohort estimates Sobiecki et al. 2023, Martínez-González et al. 2023. Polyphenol-dense green-Mediterranean variants are testing whether the pattern can be improved rather than merely exported Yaskolka Meir et al. 2021, Zelicha et al. 2022.

The soft outcomes need better trials more than more trials. Cognition and depression require studies that are longer, larger, and run in higher-risk or non-Mediterranean populations, with credible control arms — the MIND trial's active, weight-losing comparator shows how easily a strong control can erase an expected effect Barnes et al. 2023, Bizzozero-Peroni et al. 2025, Wu et al. 2025.

Conclusion

The Mediterranean diet's evidence base is unusual in nutrition: it has hard-endpoint randomised trials, they broadly agree with an enormous observational literature, and its central trial survived the most public integrity crisis in the field with its estimate intact. Taken together, the diet probably reduces major cardiovascular events by a fifth to a third against minimal intervention, probably prevents type 2 diabetes, and is consistently associated with roughly 10–25% lower mortality at high adherence — a real but modest effect that shrinks outside the Mediterranean and has never been demonstrated in a trial powered for death.

Beyond the heart and the pancreas, confidence should taper. Weight and risk-factor effects are small and partly low-certainty; the cancer association is broad but causally undemonstrated; liver, frailty, and joint findings are promising leads; depression shows a consistent direction with an untrustworthy size; and cognition is an open contradiction between what cohorts predict and what the best trial found. What would move the field is already visible: PREDIMED-Plus for hard endpoints, biomarker-measured adherence to discipline the cohorts, and properly controlled trials for the outcomes where enthusiasm currently runs ahead of proof. Until then, the defensible claim is not that this diet works wonders, but that it is one of very few dietary patterns for which "probably beneficial, at modest size, on trial evidence" can honestly be said.

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